Reconciliation to Finance

Project: Gross margin decline, Q1 → Q3 2026 · **Version:** v1.0 · **Author:** Ogunlusi Teniola Elizabeth, Data Analyst · **Date:** 17 October 2026
Reviewed by (Finance): [name] · **Review date:** [date]

---

1. Result

Q1 and Q3 2026 reconcile to Finance's control totals: net revenue ties to the decimal and gross margin to within 0.03 points, against a tolerance of 0.5% and 0.2 points. All 13 validation checks in `sql/05_validation_checks.sql` pass.

Period	My net revenue ₦M	Finance ₦M	Difference	My GM %	Finance GM %	Difference (pts)	Result
Q1 2026	5,191.2	5,191.2	0.00%	33.27	33.3	−0.03	✅
Q3 2026	3,825.6	3,825.6	0.00%	19.60	19.6	0.00	✅

The −0.03 point difference in Q1 is rounding: Finance quotes one decimal (33.3%), and the calculated figure is 33.27%.

---

2. Basis of the control totals

- **Source:** Finance, as relayed to me: *Q1 2026: ₦5,191.2M net revenue, 33.3% GM. Q3 2026: ₦3,825.6M net revenue, 19.6% GM*, on the same definitions, with the data cleaned.
- **Definitions used on both sides** (see `definitions_and_assumptions.md`): net revenue is after discounts and excludes VAT; COGS is kg × USD cost on the order date × that month's FX rate; freight, duty and warehousing are excluded; blank currency is NGN; USD lines are converted at the month's rate; only exact duplicate lines are removed.
- **Tolerance (Finance):** within about **0.5% on revenue** and **0.2 points on GM %**. Finance

also said: *if you're off, find your own gap first, then bring it to me.*

---

3. How the reconciliation was reached

Step	What happened
1. Profile and clean	Profiling found the defects in `data_quality_log.md` (the double load of `order_lines`, 94 duplicate lines, mixed dates, blank and USD currency, rep spellings). `02_clean.sql` treated them before any total was calculated.
2. First tie-out	**Q1 tied on the first attempt:** ₦5,191.2M at 33.27%.
3. Q3 did not tie	Q3 came to **₦3,565.3M at 19.30%**: revenue **6.8% short** and GM 0.3 points low, outside tolerance on both.
4. Find the gap	Following Finance's instruction, I looked for what changed between Q1 and Q3. Finance had said bag products are recorded differently at some point and asked for proof from the data. Quantities for the two bag products (P03, P16) were examined month by month.
5. Finding	Before 1 May every bag-product quantity is a multiple of 25 (kilograms); from 2 May the quantities are 1, 2, 4, 8 … 120 (bags). No overlap.
6. Fix	From 1 May, kg = qty × 25 for bag products (239 lines).
7. Re-tie	**Q3 tied exactly: ₦3,825.6M at 19.60%.** Q1 was unaffected because the change starts in May.

I found and closed the gap before bringing it to Finance. The 1 May date is proven from the data and is **not yet confirmed by the warehouse**.

---

4. What each cleaning rule is worth on the tie-out

Each rule was switched off in turn and Q1 and Q3 recalculated (queries S1 to S3 in the appendix). Every rule is needed: without any one of them the result falls outside Finance's tolerance.

Rule							

switched off	Q1 revenue ₦M	vs Finance	Q1 GM %	Q3 revenue ₦M	vs Finance	Q3 GM %	Within tolerance?
None (final)	5,191.2	0.00%	33.27	3,825.6	0.00%	19.60	✅
Bag-unit rule (×25 from 1 May)	5,191.2	0.00%	33.27	3,565.3	**−6.80%**	19.30	❌ Q3 revenue and GM
USD conversion	5,145.4	**−0.88%**	32.67	3,805.5	**−0.53%**	19.17	❌ revenue and GM, both quarters
Removal of the 94 duplicates	5,269.3	**+1.50%**	33.25	3,906.7	**+2.12%**	19.66	❌ revenue, both quarters

Not shown: leaving the `order_lines` double load in place. Every line would be counted twice, so revenue and cost would both double while the **margin % would look unchanged**. This is why row counts and naira revenue are checked as well as GM %.

---

5. Validation checks (`05_validation_checks.sql` , query 5a)

ID	Check	Expected	Actual	Status
V01	Q1 net revenue (₦M) within 0.5% of Finance	5,191.2	5,191.2	PASS
V02	Q1 gross margin % within 0.2 pts of Finance	33.3	33.27	PASS
V03	Q3 net revenue (₦M) within 0.5% of Finance	3,825.6	3,825.6	PASS
V04	Q3 gross margin % within 0.2 pts of Finance	19.6	19.60	PASS
V05	Model rows = clean lines (no join loss or multiplication)	4,712	4,712	PASS
V06	One model row per `line_id`	4,712	4,712	PASS

V07	Lines with no supplier cost or FX rate	0	0	PASS
V08	Invalid order dates	0	0	PASS
V09	Exact duplicate groups left after cleaning	0	0	PASS
V10	Clean lines whose order is missing	0	0	PASS
V11	Bag-unit break is clean (ambiguous quantities)	0	0	PASS
V12	Bridge total minus actual change in gross profit (₦)	0	0	PASS
V13	Sync-bug duplicates removed	94	94	PASS

Run record. Test run: SQLite 3.45.1, from the six source CSVs, on a single load and on a doubled load of `order_lines` : 13 of 13 PASS, identical results. My run in DBeaver (SQLite), 17 October 2026: files 02 to 05 ran without error and matched their expected values; 13 of 13 PASS. [Attach a screenshot of the 5a result.]

---

6. What is not reconciled

- **Q2 (April to June):** Finance supplied no control total. Q2 comes from the same validated pipeline but is **not independently reconciled**.
- **Months, products, customer groups and reps:** only the quarterly totals are externally reconciled. These cuts add up to the reconciled totals, but Finance has not checked them separately.
- **The margin bridge** ties to the *actual* change in gross profit (V12, a difference of ₦0). That is an internal consistency check, not an external one.
- **Gross margin excludes freight, duty and warehousing**, as in Finance's own number, so this reconciliation says nothing about contribution margin.

---

7. Sign-off

Role	Name	Date	Signature / confirmation
Prepared by (Data	Ogunlusi Teniola	17 October	

Analyst)	Elizabeth	2026	
Reviewed by (Finance)	[name]	[date]	[pending]

Appendix: sensitivity queries (section 4)

Run after files 02 to 04. These are supporting queries and are not part of the five SQL files.

```
-- S1  Without the bag-unit rule (quantity used as recorded). Expect Q1 5191.2 /
SELECT period,
       ROUND(SUM(qty * price_ngn * (1 - discount_pct / 100.0)) / 1e6, 1) AS net_r
       ROUND(100.0 * (SUM(qty * price_ngn * (1 - discount_pct / 100.0)) - SUM(qty
             / SUM(qty * price_ngn * (1 - discount_pct / 100.0)), 2) AS gm_pct
FROM margin_model WHERE period IN ('Q1','Q3') GROUP BY period ORDER BY period;

-- S2  Without converting USD lines (USD price read as naira). Expect Q1 5145.4 /
SELECT period,
       ROUND(SUM(kg * CASE WHEN cur = 'USD' THEN price_ngn / fx ELSE price_ngn EN
       ROUND(100.0 * (SUM(kg * CASE WHEN cur = 'USD' THEN price_ngn / fx ELSE pri
             / SUM(kg * CASE WHEN cur = 'USD' THEN price_ngn / fx ELSE price_ngn
FROM margin_model WHERE period IN ('Q1','Q3') GROUP BY period ORDER BY period;

-- S3  Without removing the 94 sync-bug duplicates. Expect Q1 5269.3 / 33.25, Q3
WITH dup AS (
  SELECT CASE WHEN o.order_dt BETWEEN '2026-01-01' AND '2026-03-31' THEN 'Q1'
              WHEN o.order_dt BETWEEN '2026-07-01' AND '2026-09-30' THEN 'Q3' ELS
         SUM(x.kg * x.pr * (1 - x.dp / 100.0)) AS rev,
         SUM(x.kg * x.pr * (1 - x.dp / 100.0) - x.kg * x.uc * x.fx) AS gm
  FROM (
    SELECT d.order_id, d.discount_pct AS dp,
           CASE WHEN p.unit = 'bag_25kg' AND o2.order_dt >= '2026-05-01' THEN d.q
           CASE WHEN COALESCE(NULLIF(TRIM(d.currency),''),'NGN') = 'USD'
                THEN d.unit_price * f.ngn_per_usd ELSE d.unit_price END AS pr,
           f.ngn_per_usd AS fx,
           (SELECT sc.unit_cost_usd_per_kg FROM supplier_costs sc
             WHERE sc.product_id = d.product_id AND sc.effective_date <= o2.order
             ORDER BY sc.effective_date DESC LIMIT 1) AS uc
    FROM (
      SELECT *, ROW_NUMBER() OVER (
                PARTITION BY order_id, product_id, qty, unit_price, discount_pct,
                             COALESCE(NULLIF(TRIM(currency),''),'NGN')
                ORDER BY line_id) AS rn
      FROM (SELECT DISTINCT * FROM order_lines)) d
    JOIN clean_orders o2 ON o2.order_id = d.order_id
```

```
    JOIN products p      ON p.product_id = d.product_id
    JOIN fx_rates f      ON f.month_start = o2.month
    WHERE d.rn > 1) x
  JOIN clean_orders o ON o.order_id = x.order_id
  GROUP BY 1),
m AS (SELECT period, SUM(rev) AS rev, SUM(gm) AS gm FROM margin_model GROUP BY pe
SELECT m.period,
       ROUND((m.rev + COALESCE(dup.rev, 0)) / 1e6, 1) AS net_rev_M,
       ROUND(100.0 * (m.gm + COALESCE(dup.gm, 0)) / (m.rev + COALESCE(dup.rev, 0)
FROM m LEFT JOIN dup ON dup.period = m.period
WHERE m.period IN ('Q1','Q3') ORDER BY m.period;
```